



Republic of the Philippines
North Eastern Mindanao State University

CogniSync: Addressing Philippine Student Tech Addiction Through Web-Based Generative AI Intervention

IT-related issue: Tech Addiction and Mental Health

In Partial Fulfilment of the Requirement in
CS 323 – Social Issues and Professional Practices

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Republic of the Philippines
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Activity 7

Giray, L., et al. (2024). *A survey on digital device engagement, digital stress, and coping strategies among college students in the Philippines.*

- What is the main focus of the study? The study focuses on evaluating the digital habits of Filipino university students, specifically examining the frequency of their device engagement and the resulting digital stress and coping mechanisms.
- How does the study relate to the chosen IT-related social issue? It directly addresses the core demographic of the project (Filipino college students) and establishes the behavioral baseline of technology addiction, showing how heavy reliance on digital platforms translates into physical and mental exhaustion.
- What are the key findings or conclusions of the study? The researchers found that students are heavily reliant on their devices, habitually engaging with screens immediately upon waking and right before sleeping. The study concluded that this lack of behavioral control causes severe sleep deprivation, leading to a significant plummet in students' overall academic self-efficacy.
- Are there any gaps or limitations in the study that are relevant to the project? While the study successfully identifies the causes and symptoms of digital stress (like sleep deprivation and academic decline), it primarily serves as a diagnostic survey. The gap lies in the lack of an accessible, active intervention to disrupt these habits in real-time—a gap that a localized digital wellness coach like CogniSync is designed to fill.

Lacsa, J. E. M. (2025). *The dark side of digital connectivity: Smartphone dependency and body image issues in Filipino teens.*

- What is the main focus of the study? This research investigates the psychological ramifications of smartphone dependency, specifically looking at how continuous exposure to social networking sites affects the body image and self-perception of adolescents in the Philippines.
- How does the study relate to the chosen IT-related social issue? It highlights the severe mental health consequences of tech addiction. Rather than just focusing on lost time, it connects digital overuse directly to psychological distress, which is the exact problem a digital wellness tool aims to mitigate.
- What are the key findings or conclusions of the study? The study concludes that Filipino adolescents who are deeply embedded in social media are highly vulnerable to negative self-perceptions, decreased self-esteem, body image distortion, and appearance-related anxiety due to the endless stream of idealized content.
- Are there any gaps or limitations in the study that are relevant to the project? The study documents the psychological damage but does not explore technological tools to build mental resilience against these specific triggers. This presents an opportunity to introduce Cognitive Behavioral Therapy (CBT-Lite) reframing prompts directly into the student's digital environment.



Republic of the Philippines
North Eastern Mindanao State University

Buctot, D. B., et al. (2020). *Factors associated with smartphone addiction prevalence and its predictive capacity for health-related quality of life among Filipino adolescents.*

- What is the main focus of the study? The study aims to identify the specific environmental and behavioral factors that predict the likelihood of smartphone addiction among youth in the Philippines, and how this addiction impacts their overall health-related quality of life.
- How does the study relate to the chosen IT-related social issue? It establishes the empirical evidence that smartphone addiction is a measurable, predictive public health concern in the country, reinforcing the necessity for early-stage, preventative IT solutions.
- What are the key findings or conclusions of the study? The key finding is that daily screen time volume and the user's social environment are the two most significant predictors of smartphone addiction prevalence.
- Are there any gaps or limitations in the study that are relevant to the project? The study is heavily focused on predictive modeling and relies on self-reported surveys. It does not propose a mechanism to actively alter a student's "daily screen time" or "social environment." An interactive system that provides exit strategies to analog hobbies (like local sports) directly addresses the environmental factors missing in this study.

Radtke, T., et al. (2022). *Digital detox: An effective solution in the smartphone era? A systematic literature review.*

- What is the main focus of the study? This systematic review evaluates the clinical effectiveness of various "digital detox" strategies and psychological interventions designed to combat smartphone and internet addiction.
- How does the study relate to the chosen IT-related social issue? It provides the foundational academic justification for *how* to solve the social issue. It proves that simple app-blockers are insufficient and that cognitive strategies are required for long-term habit changes.
- What are the key findings or conclusions of the study? The researchers concluded that addressing tech addiction requires a multifaceted approach. Simply removing the device is ineffective; success requires promoting digital literacy, healthy online habits, and providing robust psychological support through structured cognitive interventions.
- Are there any gaps or limitations in the study that are relevant to the project? Most of the interventions reviewed in the literature are based on Western clinical models or require formal psychiatric appointments. There is a distinct gap in providing these structured cognitive interventions through a culturally localized, easily accessible, non-personalized AI platform tailored specifically for Southeast Asian demographics.

Kemp, S. (2024). *Digital 2024: The Philippines.*

- What is the main focus of the study? The report focuses on compiling comprehensive data regarding internet, social media, and digital device usage statistics globally, with specific demographic breakdowns for countries including the Philippines.



Republic of the Philippines
North Eastern Mindanao State University

- How does the study relate to the chosen IT-related social issue? It establishes the severity of the demographic target, proving that the target audience for the CogniSync system is currently at the highest risk globally for digital overuse.
- What are the key findings or conclusions of the study? The key finding is that Filipino youth are consistently ranked among the heaviest users of the internet and social media globally.
- Are there any gaps or limitations in the study that are relevant to their project? This report provides demographic statistics and usage hours but does not offer psychological interventions or technological solutions to lower these numbers, emphasizing the need for an active tool like CogniSync.

Abad Santos, P. M. T., & Sarile, B. A. F. (2023). *The double-edged impact of Internet use on mental health outcomes among Filipino university students.*

- What is the main focus of the study? The study examines how internet use impacts the mental health outcomes of university students in the Philippines, acting as both a necessary academic tool and a detriment to well-being.
- How does the study relate to the chosen IT-related social issue? It highlights the core conflict Filipino students face—the inability to separate schoolwork from digital recreation—which is the exact behavior a digital wellness coach aims to address.
- What are the key findings or conclusions of the study? The study concludes that the increased reliance on digital platforms has blurred the lines between productive academic use and recreational overuse, elevating tech addiction to a major public health concern for Filipino students.
- Are there any gaps or limitations in the study that are relevant to their project? While it elevates the issue to a public health concern, it focuses on the "double-edged" nature of the internet rather than providing a preventative tool to help students manage that balance.

Cleofas, J. V., & Rocha, I. C. N. (2021). *Demographic, gadget and internet profiles as determinants of disease and social anxiety.*

- What is the main focus of the study? This research investigates how the recent shifts in educational environments and increased gadget profiling determine social anxiety levels among students.
- How does the study relate to the chosen IT-related social issue? It explains the environmental catalyst for the current spike in tech reliance, providing context for why students are struggling more now than in previous years.
- What are the key findings or conclusions of the study? The researchers found that the recent shift to hybrid learning environments accelerated students' increased reliance on digital platforms.
- Are there any gaps or limitations in the study that are relevant to their project? The study identifies the environmental shift to hybrid learning but does not provide the



Republic of the Philippines
North Eastern Mindanao State University

mental tools required to cope with this increased reliance, a gap that CBT-Lite prompts can fill.

Montag, C., et al. (2021). *How to overcome taxonomical problems in the study of Internet use disorders and what to do with "smartphone addiction"?*

- What is the main focus of the study? The study focuses on the clinical and taxonomical definitions of internet use disorders, arguing for the formal recognition of smartphone and internet overuse as behavioral addictions.
- How does the study relate to the chosen IT-related social issue? It provides the clinical validation for the project's domain limitation, proving that "tech addiction" is a legitimate, non-substance behavioral issue requiring intervention.
- What are the key findings or conclusions of the study? The study concludes that internet and smartphone addiction are increasingly recognized within clinical literature as non-substance behavioral addictions characterized by an uncontrollable, impulsive need to connect to digital networks.
- Are there any gaps or limitations in the study that are relevant to their project? The paper focuses heavily on clinical taxonomy and definitions for researchers. It does not provide student-facing preventative measures for those who are "at-risk" but not yet clinically diagnosed.

Elhai, J. D., et al. (2020). *Depression and anxiety symptoms are related to problematic smartphone use severity... Fear of missing out as a mediator.*

- What is the main focus of the study? This research examines the relationship between problematic smartphone use and symptoms of anxiety and depression, specifically analyzing how the "Fear of Missing Out" (FOMO) mediates this relationship.
- How does the study relate to the chosen IT-related social issue? It identifies the specific psychological mechanism (FOMO) that drives the compulsion to scroll, which is vital for programming an AI to recognize and respond to user symptoms accurately.
- What are the key findings or conclusions of the study? The researchers identified that FOMO on virtual social validation is globally recognized as a primary mediator between problematic smartphone use and anxiety.
- Are there any gaps or limitations in the study that are relevant to their project? The study successfully identifies FOMO as the primary mediator but does not implement a real-time intervention system to disrupt the FOMO response when it occurs, creating an opportunity for localized AI coaching.



Republic of the Philippines
North Eastern Mindanao State University

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